Relation-oversampling pilot — run-20261111T0451Z

Status: pilot under a DRAFT protocol. Not the measurement; supports no claim. All numbers are `[pcc:unverified]`.

Outcome and checks

- 4 cells × 5 seeds × 6 arms = 120 rows requested, 120 completed, **0 failed verification** (every arm re-checked `[log]G == point` for every column).
- In every cell, every oversampled arm (f = 0.1, 0.25, 0.5) and the `ic logs` reference arm had a **higher** total counted cost than f = 0, at $\lambda = 0$, $\hat{\chi}/10$, $\hat{\chi}$ and $10\hat{\chi}$. The extra collection trials dominate; LA row operations also rose on average in 15 of 16 cell–arm pairs (exception: C4 `logs_default`, −523 row_ops mean, which did not offset its +42.8 trials).
- C1–C3 (CADO filter) leave a Krylov core of 0–1; C4 (legacy filter) leaves 8–11.

What ran

- Command: `research/logs_oversampling_20261111/pilot/run_pilot.sh` under bench record `benchmarks/records/20261111T0443Z-logs-oversampling-pilot.json` (record `commit` field 6df0f91 is from `init`; the run used checkout 9373a70, which differs only in the run script and factor-base spec files; Rust identical to a4867885, the commit CI tested).
- Binary: `target/release/ic` (`cargo build --release --bin ic`), `ic oversample --seeds 5 --fractions 0,0.1,0.25,0.5 --aa-control --logs-arm`, `RAYON_NUM_THREADS=1`.
- Host: shared cloud container, Intel Xeon @ 2.30 GHz, 4 CPUs, isolation **unstated** (no L2). Wall times are indicative only; counted units (trials, row_ops) are exact. Local route at the owner's request; S3 resume checkpoint waived by the owner (Git instead).
- Cells: C1 deg 23 m=2 factor-index 0; C2 deg 37 m=3 `ic search` selection (1184 pts); C3 deg 37 m=3 subgroup_orbits 7400 (CADO); C4 = C3 with legacy filter. Seeds: base `0x4b6f626c69747a00` + i, i = 0..4.

Results (derived by `pilot/analyze.py` from `pilot/rows/*.jsonl`)

C(f) = trials + λ·row_ops. $\hat{\chi}$ = median over f0 arms of (LA s / row_op) ÷ (collection s / trial). ΔX = arm − f0 on the same seed's pool. CI: paired bootstrap over 5 seeds, 10,000 resamples.

cell	cols	failed	$\hat{\chi}$ (trials/row-op)	mean C(f0) at $\hat{\chi}$	f0 core dim (5 seeds)	LA share of f0 wall	A/A max |ΔLA| (ms)
C1	44	0/30	0.142	395.3	0,0,0,0,0	17.46%	0.133
C2	16	0/30	0.0144	29.3	0,0,0,0,0	8.28%	0.059
C3	104	0/30	0.00561	314.9	1,1,1,0,1	4.26%	0.531
C4	104	0/30	0.000253	305.1	10,10,8,8,11	0.69%	0.153

cell	arm	Δtrials	Δrow_ops	ΔC at $\hat{\chi}$, 95% CI	ΔC at $10\hat{\chi}$	Δwall (ms)
C1	f0.1	16.2	11.6	17.9 [11.8, 26.1]	32.7	0.31
C1	f0.25	31.6	20.2	34.5 [27.7, 45.3]	60.3	0.72
C1	f0.5	61.8	33.8	66.6 [56.8, 80.1]	109.9	1.47
C1	logs_default	32.4	25.8	36.1 [14.2, 58.0]	69.1	0.59
C2	f0.1	3.6	36.4	4.1 [3.3, 4.9]	8.8	0.11
C2	f0.25	6.2	68.2	7.2 [6.6, 7.8]	16.0	0.20
C2	f0.5	11.2	123.4	13.0 [11.7, 14.9]	28.9	0.37
C2	logs_default	36.2	281.0	40.2 [34.4, 46.1]	76.5	1.20
C3	f0.1	11.0	8.4	11.0 [10.0, 11.8]	11.5	2.09
C3	f0.25	26.0	29.0	26.2 [25.5, 26.8]	27.6	4.71
C3	f0.5	52.0	123.4	52.7 [52.1, 53.4]	58.9	9.37
C3	logs_default	42.8	25.4	42.9 [22.4, 60.5]	44.2	7.74
C4	f0.1	11.0	103.6	11.0 [10.7, 11.3]	11.3	1.94
C4	f0.25	26.0	226.6	26.1 [25.7, 26.4]	26.6	4.50
C4	f0.5	52.0	107.2	52.0 [51.8, 52.3]	52.3	9.05
C4	logs_default	42.8	−523.2	42.7 [21.2, 60.5]	41.5	7.49

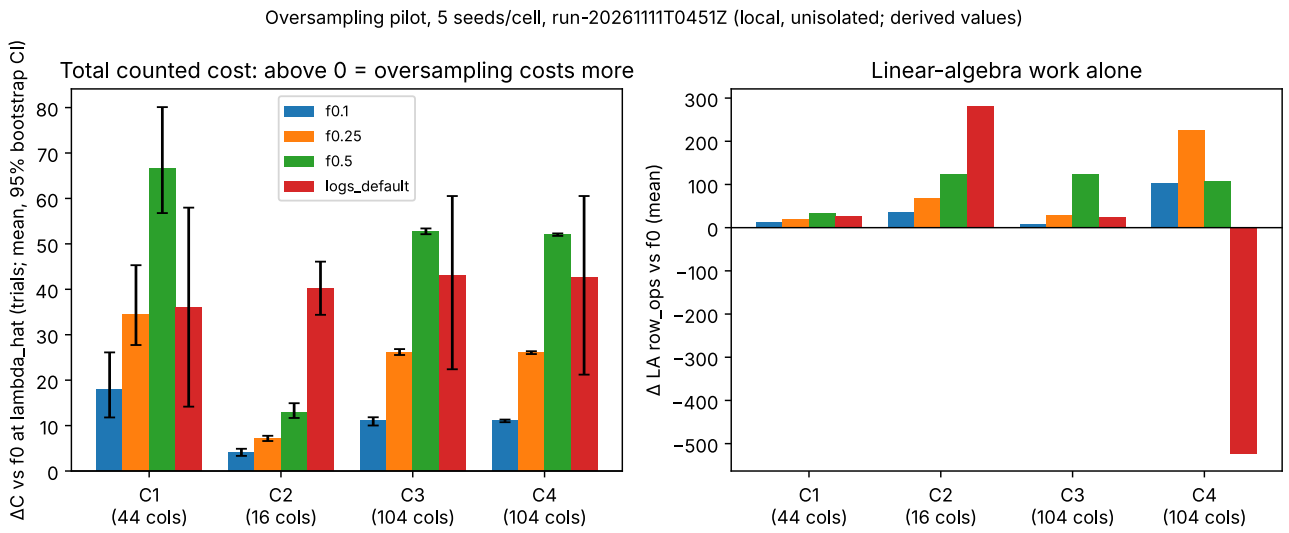


Figure 1: Paired differences vs f0 per cell, 5 seeds each, run-20261111T0451Z. Left: total counted cost at $\hat{\lambda}$ (mean, 95% bootstrap CI). Right: LA row operations (mean). Derived values.

Sizing the frozen run

`pilot_summary.json` gives, per arm and λ, the seeds needed for a 95% CI half-width of 1% of mean C(f0): ≤ 60 for all f arms except C2 f0.5 (207), where C(f0) is only 29 trials. For deciding the **sign** of ΔC the 5-seed intervals already exclude 0 in every cell. Proposed `seeds_per_cell`: 30 (unchanged) — a decision for the protocol reviewer at freeze.

Raw records

`research/logs_oversampling_20261111/pilot/rows/C{1..4}.jsonl` (rows) and `C{1..4}.out.json` (command output); summary `pilot/pilot_summary.json`. Raw files are hashed in the bench record; S3 upload not possible from this container (no AWS identity).